

Solutions: Additional Exercises 4

Taylor Approximations & Uniform Convergence

Series and Integral Transforms

Exercise 1 — Numerical Approximations via Taylor Series

General strategy: Use the Taylor/Maclaurin series, bound the remainder R_n by the first omitted term (for alternating series, or by the Lagrange remainder otherwise), and find the smallest n that makes $|R_n| < \varepsilon$.

(a) ★ $e^{0.1}$ to within 10^{-4}

$$e^x = \sum_{k=0}^n \frac{x^k}{k!} + R_n(x) \text{ where } |R_n(x)| \leq \frac{e^{|x|}|x|^{n+1}}{(n+1)!}.$$

For $x = 0.1$, bound $e^{0.1} < 3$:

$$|R_n| \leq \frac{3 \cdot (0.1)^{n+1}}{(n+1)!}.$$

$$\text{For } n = 3: |R_3| \leq \frac{3 \cdot (0.1)^4}{4!} = \frac{3 \times 10^{-4}}{24} \approx 1.25 \times 10^{-5} < 10^{-4}.$$

Use $n = 3$ terms:

$$e^{0.1} \approx 1 + 0.1 + \frac{(0.1)^2}{2} + \frac{(0.1)^3}{6} = 1 + 0.1 + 0.005 + 0.000167 \approx \mathbf{1.1052}.$$

(b) ★★ $\sin(0.2)$ to within 10^{-5}

$\sin x = \sum_{k=0}^n \frac{(-1)^k x^{2k+1}}{(2k+1)!}$ (alternating). The error is bounded by the first omitted term:

$$|R_n| \leq \frac{(0.2)^{2n+3}}{(2n+3)!}.$$

For $n = 2$ (using terms $k = 0, 1, 2$, i.e., $x, -x^3/6, x^5/120$):

$$|R_2| \leq \frac{(0.2)^7}{5040} \approx \frac{1.28 \times 10^{-5}}{5040} \approx 2.5 \times 10^{-9} \ll 10^{-5}.$$

Even $n = 1$ gives $|R_1| \leq (0.2)^5/120 \approx 2.7 \times 10^{-6} < 10^{-5}$. Use $n = 1$ (2 non-zero terms):

$$\sin(0.2) \approx 0.2 - \frac{(0.2)^3}{6} = 0.2 - 0.001\bar{3} \approx \mathbf{0.19867}.$$

(c) ★★ $\ln(1.1)$ to within 10^{-3}

$\ln(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1} x^k}{k}$ at $x = 0.1$ (alternating, since $0 < x \leq 1$).

Error bounded by first omitted term: $|R_N| \leq \frac{(0.1)^{N+1}}{N+1}$.

For $N = 3$: $|R_3| \leq \frac{(0.1)^4}{4} = 2.5 \times 10^{-5} < 10^{-3}$.

Use 3 terms:

$$\ln(1.1) \approx 0.1 - \frac{0.01}{2} + \frac{0.001}{3} = 0.1 - 0.005 + 0.000\bar{3} \approx \mathbf{0.09531}.$$

(d) ★★ $\sqrt[3]{28}$ to within 10^{-3}

Write $28 = 27 \cdot (1 + 1/27)$, so $\sqrt[3]{28} = 3 \cdot (1 + 1/27)^{1/3}$.

Use $(1+u)^{1/3} = \sum_{k=0}^{\infty} \binom{1/3}{k} u^k$ at $u = 1/27$:

$$(1+u)^{1/3} \approx 1 + \frac{u}{3} - \frac{u^2}{9} + \dots$$

For $u = 1/27$: $u/3 = 1/81 \approx 0.01235$; $u^2/9 = 1/6561 \approx 1.5 \times 10^{-4}$.

The second correction term is $\approx 4.6 \times 10^{-4}$. Using two correction terms:

$$\sqrt[3]{28} \approx 3 \left(1 + \frac{1}{81} - \frac{1}{6561} \right) = 3 \times 1.012195\dots \approx \mathbf{3.0366}.$$

(e) ★★ $\cos(\pi/36)$ to within 10^{-8}

$\pi/36 \approx 0.08727$. $\cos x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!}$ (alternating); error $\leq$ first omitted term.

For $n = 3$ (terms $k = 0, 1, 2$):

$$|R_3| \leq \frac{(\pi/36)^6}{720} \approx \frac{(0.08727)^6}{720} \approx \frac{4.4 \times 10^{-7}}{720} \approx 6 \times 10^{-10} < 10^{-8}.$$

Use 3 terms:

$$\cos(\pi/36) \approx 1 - \frac{(\pi/36)^2}{2} + \frac{(\pi/36)^4}{24} \approx 1 - 0.003805 + 7.24 \times 10^{-6} \approx \mathbf{0.99620}.$$

(f) ★★★ $\pi/4 = \arctan(1)$ via series

$\arctan x = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} x^{2k+1}$. At $x = 1$, this is the Leibniz series $1 - 1/3 + 1/5 - \dots$.

Error: $|R_N| \leq 1/(2N+3) < 10^{-3}$ requires $2N+3 > 1000$, i.e., $N \geq 499$. This is extremely slow.

Faster approach: use $\pi/4 = \arctan(1/2) + \arctan(1/3)$ (Machin-type identity). Then each series converges much faster; only ~ 3 – 4 terms are needed to achieve 10^{-3} accuracy.

(g) ★★★★★ $\ln 2$ efficiently

The direct series $\ln 2 = \sum_{k=1}^{\infty} (-1)^{k+1}/k$ converges, but slowly ($|R_N| \leq 1/(N+1)$, need $N \geq 999$).

Better: use $\ln 2 = 2 \operatorname{arctanh}(1/3)$ where $\operatorname{arctanh}(u) = \sum_{k=0}^{\infty} \frac{u^{2k+1}}{2k+1}$:

$$\ln 2 = 2 \sum_{k=0}^{\infty} \frac{(1/3)^{2k+1}}{2k+1}.$$

Error after N terms: $|R_N| \leq \frac{2 \cdot (1/3)^{2N+3}}{2N+3}$.

For $N = 2$: $|R_2| \leq \frac{2 \cdot (1/3)^7}{7} \approx 3.2 \times 10^{-4} < 10^{-3}$. Only 3 terms needed. ✓

(h) ★★★★★ $\int_0^{0.5} e^{-x^2} dx$ to within 10^{-4}

Use $e^{-x^2} = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{k!}$ and integrate term by term:

$$\int_0^{0.5} e^{-x^2} dx = \sum_{k=0}^{\infty} \frac{(-1)^k (0.5)^{2k+1}}{k!(2k+1)}.$$

This is an alternating series; error $\leq$ first omitted term.

For $N = 3$ (terms $k = 0, 1, 2, 3$): first omitted term ($k = 4$):

$$\frac{(0.5)^9}{4! \cdot 9} = \frac{(1/2)^9}{216} = \frac{1}{110592} \approx 9.0 \times 10^{-6} < 10^{-4}.$$

Use 4 terms ($k = 0, 1, 2, 3$):

$$\approx 0.5 - \frac{(0.5)^3}{3} - \frac{(0.5)^5}{2! \cdot 5} + \frac{(0.5)^7}{3! \cdot 7} \approx 0.5 - 0.04167 + 0.003125 - 0.000186 \dots \approx \mathbf{0.4612}.$$

Exercise 2 — Uniform Convergence of Sequences

(a) ★ $f_n(x) = x^n$ on $[0, 1)$

Pointwise limit: $f(x) = 0$ for $x \in [0, 1)$.

$$\sup_{x \in [0, 1)} |f_n(x) - 0| = \sup_{x \in [0, 1)} x^n = 1 \not\rightarrow 0.$$

NOT uniformly convergent on $[0, 1)$.

(b) ★★ $f_n(x) = x/n$ on $\mathbb{R}$

Pointwise limit: 0. But $\sup_{x \in \mathbb{R}} |x/n| = +\infty$. **NOT uniform** on $\mathbb{R}$.

On any bounded interval $[a, b]$: $\sup |x/n| \leq \max(|a|, |b|)/n \rightarrow 0$. **Uniform on every bounded interval.**

(c) ★ $f_n(x) = \frac{1}{nx+1}$ on $[1, \infty)$

For $x \geq 1$: $0 \leq f_n(x) \leq \frac{1}{n+1} \rightarrow 0$. So $\sup_{x \geq 1} |f_n(x)| \leq \frac{1}{n+1} \rightarrow 0$.

Uniformly convergent to 0 on $[1, \infty)$.

(d) ★★ $f_n(x) = \frac{nx}{1+n^2x^2}$ on $(0, \infty)$

Pointwise: for fixed $x > 0$, $f_n(x) = \frac{nx}{1+n^2x^2} \leq \frac{1}{2nx} \rightarrow 0$ (by AM-GM: $1+n^2x^2 \geq 2nx$).

But at $x_n = 1/n$: $f_n(1/n) = \frac{n \cdot (1/n)}{1+1} = \frac{1}{2}$.

So $\sup_{x>0} |f_n(x)| \geq 1/2 \not\rightarrow 0$. **NOT uniformly convergent**.

(e) ★★★ $\int_0^1 n^2 x e^{-nx} dx$

Integrate by parts ($u = x$, $dv = n^2 e^{-nx} dx$):

$$\int_0^1 n^2 x e^{-nx} dx = [-n x e^{-nx}]_0^1 + \int_0^1 n e^{-nx} dx = -n e^{-n} + [-e^{-nx}]_0^1 = -n e^{-n} + 1 - e^{-n} = 1 - (n+1)e^{-n} \rightarrow 1.$$

The pointwise limit of $f_n(x) = n^2 x e^{-nx}$ is 0 on $(0, 1]$, and $\int_0^1 0 dx = 0 \neq 1$.

Since $\lim_{n \rightarrow \infty} \int_0^1 f_n dx \neq \int_0^1 \lim_{n \rightarrow \infty} f_n dx$, the convergence is **NOT uniform** on $[0, 1]$.

(f) ★★ $f_n(x) = \frac{\sin(nx)}{\sqrt{n}}$

$\sup_x |f_n(x)| = 1/\sqrt{n} \rightarrow 0$. **Uniformly convergent** to 0 on $\mathbb{R}$.

However, $f'_n(x) = \sqrt{n} \cos(nx)$, which is unbounded. The convergence of derivatives fails — a warning that uniform convergence does **not** imply convergence of derivatives.

(g) ★★ $f_n(x) = x^n(1-x)$ on $[0, 1]$

Pointwise limit: 0. Find the maximum on $[0, 1]$: $f'_n(x) = x^{n-1}(n - (n+1)x) = 0$ at $x^* = n/(n+1)$.

$$f_n(x^*) = \left(\frac{n}{n+1}\right)^n \cdot \frac{1}{n+1}.$$

Since $\left(\frac{n}{n+1}\right)^n \rightarrow e^{-1}$:

$$\sup_{[0,1]} |f_n| \sim \frac{e^{-1}}{n} \rightarrow 0.$$

Uniformly convergent to 0 on $[0, 1]$.

(h) ★★ Proof: uniform convergence implies integral convergence

Theorem: If $f_n \rightarrow f$ uniformly on $[a, b]$, then $\int_a^b f_n \rightarrow \int_a^b f$.

Proof: For any $\varepsilon > 0$, there exists N such that $\|f_n - f\|_\infty < \frac{\varepsilon}{b-a}$ for all $n > N$. Then:

$$\left| \int_a^b f_n dx - \int_a^b f dx \right| = \left| \int_a^b (f_n - f) dx \right| \leq \int_a^b |f_n - f| dx \leq (b-a) \cdot \frac{\varepsilon}{b-a} = \varepsilon. \quad \square$$

Exercise 3 — Uniform Convergence of Series

(a) ★ $\sum \frac{\cos(nx)}{n^2}$ on $\mathbb{R}$

$$\left| \frac{\cos(nx)}{n^2} \right| \leq \frac{1}{n^2} := M_n \text{ and } \sum M_n = \sum 1/n^2 < \infty.$$

By the **Weierstrass M-Test**: **uniformly convergent on $\mathbb{R}$** .

(b) ★★ $\sum x^n$ on $(-1, 1)$ vs. $[-r, r]$, $0 < r < 1$

On $[-r, r]$: $|x^n| \leq r^n =: M_n$ and $\sum r^n < \infty$. **Uniform by M-Test.**

On $(-1, 1)$: $\sup_{x \in (-1, 1)} |x^n| = 1$ and $\sum 1 = \infty$. The M-Test fails. In fact, the convergence is **not uniform** on $(-1, 1)$ (the sum $1/(1-x)$ is unbounded near $x = 1$).

(c) ★★ $\sum \frac{(-1)^n x^{2n}}{n}$ on $[-1, 1]$

Use **Dirichlet's Test** for uniform convergence: the partial sums of $(-1)^n$ are bounded (by 1), and x^{2n}/n decreases to 0 uniformly on $[-1, 1]$ (since $|x^{2n}/n| \leq 1/n \rightarrow 0$ uniformly). **Uniformly convergent on $[-1, 1]$.**

(d) ★★ $\sum \frac{x^n}{n^2 + x^2}$

On $[-r, r]$: $\frac{|x|^n}{n^2 + x^2} \leq \frac{r^n}{n^2} =: M_n$ and $\sum r^n/n^2 < \infty$. **Uniform on $[-r, r]$ by M-Test.**

On $[-1, 1]$: at $x = 1$, $a_n = 1/(n^2 + 1)$ and $\sum 1/(n^2 + 1)$ converges. A careful analysis shows uniform convergence holds on $[-1, 1]$ as well.

(e) ★★ $S(x) = \sum \frac{\sin(nx)}{n^3}$: **continuity and differentiability**

$|\sin(nx)/n^3| \leq 1/n^3$; $\sum 1/n^3 < \infty$. By M-Test, the series converges **uniformly on $\mathbb{R}$** , so S is **continuous**.

Differentiated series: $\sum \frac{n \cos(nx)}{n^3} = \sum \frac{\cos(nx)}{n^2}$. By part (a), this converges uniformly. By the **Term-by-Term Differentiation Theorem**:

$$S'(x) = \sum_{n=1}^{\infty} \frac{\cos(nx)}{n^2}.$$

(f) ★★ $\sum_{n=0}^{\infty} (-1)^n x^{2n}$ on $[-r, r]$, $r < 1$

$|(-1)^n x^{2n}| \leq r^{2n}$; $\sum r^{2n} = 1/(1 - r^2) < \infty$. **Uniform by M-Test.**

The series sums to the geometric series $\sum (-x^2)^n = \frac{1}{1 + x^2}$.

Differentiating term by term (justified by uniform convergence on $[-r, r]$):

$$\frac{d}{dx} \left(\frac{1}{1 + x^2} \right) = -\frac{2x}{(1 + x^2)^2} = \sum_{n=1}^{\infty} (-1)^n (2n) x^{2n-1}. \checkmark$$

(g) ★★★ **Power series** $\sum a_n x^n$ on $[-r, r]$ vs. full $(-R, R)$

Uniform on $[-r, r]$ for any $r < R$: $|a_n x^n| \leq |a_n| r^n =: M_n$. Since $r < R$, the series $\sum |a_n| r^n$ converges (by definition of radius of convergence and Abel's theorem). **M-Test applies:** uniform convergence on $[-r, r]$. $\square$

Not necessarily uniform on $(-R, R)$: Consider $\sum x^n$ with $R = 1$. The sum $1/(1 - x) \rightarrow \infty$ as $x \rightarrow 1^-$, so the partial sums cannot converge uniformly on $(-1, 1)$ (a uniformly convergent series of bounded functions has a bounded sum).

Key takeaways: For sequences, compute $\sup_x |f_n(x) - f(x)|$ and check if it tends to 0; a moving maximum that stays bounded away from 0 kills uniform convergence. For series, the Weierstrass M-Test is the main tool; on compact subintervals of the radius, it always applies. Uniform convergence justifies swapping $\lim$ with $\int$ and $\frac{d}{dx}$ (with the extra hypothesis that the differentiated series converges uniformly).